DHRUV DUBEY

AI Engineer | Agentic AI, LLM Systems & RAG | College Park, MD
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EDUCATION

University of Maryland, College Park — M.S., Data Science (GPA 4.00/4.00) *Aug 2025 – May 2027*

MIT-ADT University — B.Tech, Computer Science & Engineering (GPA 3.33/4.00) *2021 – 2025*

TECHNICAL SKILLS

Languages: Python, SQL, C++, R
Agents & LLMs: Agentic AI, Multi-Agent Systems, LangGraph, LangChain, CrewAI, Model Context Protocol (MCP), Tool Calling, Prompt & Context Engineering, RAG, Vector Databases, Embeddings, LoRA/QLoRA Fine-Tuning, Open-Weight LLMs (Qwen3 via Ollama), Snowflake Cortex
Evaluation: LLM & Agent Evaluation, Golden Datasets, CI Eval Gates, Crash-Recovery Benchmarks, Regression Testing
Backend & MLOps: FastAPI, REST APIs, Pydantic, Microservices, Docker, Kubernetes (HPA, Probes), GitHub Actions CI/CD, MLflow, AWS, SLURM/HPC
ML: PyTorch, TensorFlow, Scikit-learn, Deep Learning, NLP, Computer Vision, Time Series, Anomaly Detection
Data: Snowflake, PostgreSQL, SQLite, Redis, MongoDB, Neo4j, Supabase, Power BI, Pandas, NumPy

EXPERIENCE

Allegis Global Solutions — AI Automation Engineer Intern *Jun 2026 – Sep 2026*
- Built a multi-agent triage system on Snowflake Cortex (Agents, Search, Analyst) that traces pipeline failures through a dependency graph to the broken stored procedure; root-cause diagnosis ~65% faster (weeks to days).
- Engineered semantic views and a YAML instruction layer from prior tickets so the agent handles new failure patterns without retraining.
- Deployed it into analysts' Power BI workflow as a chat interface, cutting fix-proposal time from hours to minutes.

University of Maryland — Research Intern, Neuro-Symbolic AI *Sep 2025 – Sep 2026*
- Co-developed a six-stage reproducibility audit of neuro-symbolic AI: screened 5,497 records to 1,304 studies and reproduced 85 codebases on the Zaratan HPC cluster (SLURM, Docker).
- Found inconsistencies in setups, hyperparameters and evaluation methods across reasoning frameworks; co-authored the paper, now under review at IEEE TNNLS.

Fortytwo Labs — Machine Learning Intern *Feb 2025 – Jul 2025*
- Built TensorFlow autoencoders on login behavior (+40% anomaly detection accuracy) and KMeans/DBSCAN clustering that surfaced 35% more outliers.
- Integrated the models with the Drools rule engine for automated risk scoring, cutting manual security reviews 33%.

Intrainz — Data Science Intern *May 2023 – Jul 2023*
- Built a crop recommendation system on 5,000+ records, raising decision-support precision 15% across 150 pilot farmers.

PROJECTS

Pipeline Incident Triage Graph GitHub
LangGraph, Model Context Protocol, FastAPI, Pydantic, Docker, Kubernetes, GitHub Actions, pytest
- Open-source multi-agent triage: a tracer walks the pipeline DAG upstream from the symptom, then specialist and reviewer agents draft and check a fix behind a human-approval gate.
- CI gated on two 100-incident eval sets: root-cause accuracy 1.00; noisy logs drop category accuracy to 0.81, which the gate tracks. Ships an MCP server and Kubernetes manifests.

Agent Harness Runtime GitHub
Python, FastAPI, SQLite, Model Context Protocol, Pydantic, Docker, Kubernetes, GitHub Actions
- Runtime owning the loop, context and per-agent sandbox, with a crash-safe run log; drives two local Qwen3 8B agents: a read-only diagnoser (28 of 40 root causes) and a fixer the sandbox holds to the human-approved plan.
- 200-run crash benchmark: 0 duplicated side effects vs 145 on restart; with the triage agent inside it, 75 fixes killed mid-apply gave 0 duplicated vs 61.
- Added a loop that tunes it against its own benchmark: human interruptions after a crash 30 of 75 to 0, nothing duplicated or dropped, 4 of 10 candidates rejected by a safety gate.

Krishi Mitra — Multilingual Agricultural RAG Assistant GitHub
- Fine-tuned LLaMA 3.2 3B with RAG answering farmers in Hindi, Marathi and English; +20% query-resolution accuracy across a 300-farmer pilot.

Smart Plant Health Monitor GitHub
- YOLOv5, ResNet and LSTM system classifying plant health at 90% accuracy, served as a Dockerized FastAPI microservice.

PUBLICATIONS

- Neuro-Symbolic AI Reproducibility Review — University of Maryland; arXiv:2608.26236; under review, IEEE TNNLS.
- Krishi Mitra: LLM-Driven QA System for Indian Farmers — MIT-ADT University.